Sara Stephens

Curriculum Vitae

Ithaca, NY USA
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Education

2021 – 2027 **Ph.D. in Mathematics**, *Cornell University*, Ithaca, NY
Advisor: Daniel Halpern-Leistner
Mathematics MS 2024

2017 – 2021 **BScH in Mathematics**, *Queen's University*, Kingston, ON, Canada
Major in Mathematics with Distinction

Research

Algebraic geometry, moduli theory, algebraic stacks, and enumerative invariants.

2026 **Alternative Moduli of Twisted Curves**, with Daniel Halpern-Leistner, Andres Fernandez Herrero and Giovanni Inchiostro.
Available at arXiv:2610.30069.

2026 **Strictly Semistable Quasimaps on the Wall**
Available at arXiv:2606.13783.

Awards & Scholarships

2025 – 2026 **Cornell Graduate School Travel Grants**
$1400 USD of funding for two international conferences.

2023 – 2026 **NSERC Alexander Graham Bell Canada Graduate Scholarship-Doctoral**
$105,000 CAD of funding for three years. Declined due to pursuing studies outside of Canada.

2023 – 2026 **NSERC Postgraduate Scholarship - Doctoral**
$63,000 CAD of funding for three years. Accepted.

2023 **AWM Emerging Mathematicians Travel Grant**
$500 USD from the Cornell Chapter of the Association for Women in Mathematics (AWM).

2022 **Cornell University Mathematics Summer Research Fellowship**

2021 **Cornell University Mathematics First Year Fellowship**
Funding for first year in graduate school by the Cornell math department.

2021 **Sports Analytics Research Fellowship**
Funding to conduct research on *Geometrically Modelling Basketball* with Dr. Catherine Pfaff.

2020 **NSERC Undergraduate Student Research Award**
Funding to conduct research on *Invariant Theory of Reflection Groups* with Dr. David Wehlau.

2020 **Queen's University Nellie and Ralph Jeffery Award in Mathematics**
Awarded to a final year mathematics major having the highest standing in mathematics courses.

2019 **Queen's University Appeal Undergraduate Scholarship**
Awarded to an upper-year student with high academic standing in Arts and Science.

Talks & Presentations

Dec 2026 **Canadian Mathematical Society Winter Meeting**, *Commutative Algebra Session.*
20 minute talk, "Alternative Moduli of Curves."

Dec 2026 **University of Washington Algebra and Algebraic Geometry Seminar.**
"Strictly Semistable Quasimaps on the Wall."

Dec 2026 **University of Pennsylvania Math-Physics Seminar.**
"Strictly Semistable Quasimaps on the Wall."

Nov 2026 **Algebraic Geometry Northeastern Series at Brown University.**
Poster, "Strictly Semistable Quasimaps on the Wall."

Nov 2026 **Queen's University Algebra & Geometry Seminar.**
"Alternative Moduli of Curves."

Oct 2026 **Columbia University Algebraic Geometry Seminar.**
"Strictly Semistable Quasimaps on the Wall."

Sep 2026 **Route 81 Conference on Commutative Algebra and Algebraic Geometry.**
40 minute talk, "Strictly Semistable Quasimaps on the Wall."

Sep 2026 **Cornell University Algebraic Geometry Seminar.**
"Strictly Semistable Quasimaps on the Wall."

Jul 2026 **GAeL XXXIII.**
30 minute talk, "Strictly Semistable Quasimaps on the Wall."

Apr 2026 **GAPXX—Kyoto.**
Poster, "Wall-Crossing for Quasimaps to $\mathbb{P}^N$."

Jan 2026 **Banff International Research Station**, *Geometry of Algebraic Stacks Workshop.*
10 minute talk, "Strictly Semistable Quasimaps to $\mathbb{P}^N$."

Dec 2025 **Canadian Mathematical Society Winter Meeting**, *Algebraic Geometry Session.*
20 minute talk, "Strictly Semistable Quasimaps to $\mathbb{P}^N$."

Teaching Experience

Cornell University

Spring 2026 **MATH 1340: Strategy, Cooperation, and Conflict**, Teaching Assistant

Fall 2025 **MATH 1110: Calculus I**, Instructor

Spring 2023 **MATH 2940: Linear Algebra for Engineers**, Teaching Assistant

Fall 2023 **MATH 2940: Linear Algebra for Engineers**, Teaching Assistant

K-12 Outreach

Spring 2025 **Ithaca High School**, Math Seminar Instructor
Designed and taught twenty lessons on the geometry of plane curves for 10th–12th grade.

Spring 2024 **Cornell Graduate Student School Outreach Program**, Teaching Fellow
Taught five lessons on projective geometry for 6th grade in a local elementary school.

Fall 2023 **Math Explorers' Club**, Instructor
Taught two interactive sessions exploring geometry and set theory for 8th–12th grade.

Queen's University

Fall 2020 **Math 110: Linear Algebra**, Undergraduate Teaching Assistant

Spring 2021 **Math 110: Linear Algebra**, Undergraduate Teaching Assistant

Service

2027 **AMS Spring Eastern Sectional Meeting**, Special Session Organizer
Co-organizing the special session "Moduli Spaces and Algebraic Stacks."

2025-Present **Cornell Algebraic Geometry Seminar**, Organizer
Co-organized Cornell's weekly algebraic geometry seminar with faculty.

2025-2026 **Cornell Mathematics TA Training**, Facilitator
Facilitated sessions on professionalism, roles and responsibilities of a TA, and sample recitations.

2022-2026 **Cornell AWM**, Mentor
Mentored four female Cornell undergraduates interested in pursuing mathematics.

2026 **AWM Introduction to LaTex Workshop**, Organizer
Organized two introduction to LaTeX workshops for STEM undergraduates at Cornell.

2023 **Olivetti Club**, Organizer
Co-organized Cornell's weekly math graduate student seminar.

2022-2023 **Directed Reading Program**, Mentor
Mentored four Cornell undergraduates in various reading projects.

2022-2023 **Expanding Your Horizons**, Workshop Leader
Co-organized two full day workshops introducing middle school girls to topology, probability and game theory.

2022 **Julia Robinson Math Festival**, Organizer
Co-organized a virtual workshop exploring mathematical puzzles for 3rd-6th grade.

2022 **Graduate Mentorship Program**, Coordinator
Co-organized Cornell's mentorship program for incoming math graduate students.

Professional Development

Fall 2026 **Teaching Certificate**, Cornell Mathematics Department
Completed Cornell's mathematics instructor development program, including giving a teaching seminar talk as well as faculty and peer observations.

Spring 2025 **Essentials of Teaching II Institute**, Cornell Center for Teaching Innovation
A four-session course covering active learning, responsible AI use, and collaborative learning, ending in a capstone activity to design active learning materials.